

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Title – **NAAC Accreditation Multi-Agent Assistant**

Domain of Project – **Education/Institutional Governance**

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Problem Statement

The Challenge

Higher education institutions undergoing NAAC (National Assessment and Accreditation Council) accreditation struggle with massive volumes of unstructured institutional data. Compiling Self-Study Reports (SSR) requires precise adherence to specific criteria metrics (e.g., tracking curriculum revisions, syllabus consistency, and academic calendars). Information is typically scattered across multiple department silos, making manual verification tedious, prone to compliance errors, and highly time-consuming.

The Objective

Build an AI-powered institutional compliance agent that automates data metrics structure parsing and drafts narrative descriptions for NAAC Criteria reporting. The solution must:

Extract Data Metrics: Auto-structure mandatory compliance data (e.g., identifying boolean statuses for academic calendars, lists of syllabus revisions, and total counts).

Knowledge Retrieval (RAG): Fetch guidelines from official NAAC institutional manuals to eliminate compliance hallucination.

Multi-Agent Coordination: Orchestrate dedicated steps to process queries, validate metric schemas, and draft high-quality narrative summaries.

Proposed Solution

The proposed solution is an AI-Powered NAAC Accreditation Assistant built using a combination of Streamlit, LangFlow, and ibm-watsonx-ai infrastructure running specialized Llama/Granite language models.

The core system utilizes a secure API backend to process complex multi-agent workflows. When an administrator or team member inputs a prompt requesting specific report parameters, the architecture triggers a modular processing sequence:

RAG Knowledge Fetcher Agent: References vector database stores containing up-to-date criteria compliance manuals.

Schema & Metric Validator Agent: Formats strict programmatic outputs (such as JSON or Python dictionaries mapping mandatory counts, boolean flags, and revision lists).

Narrative Synthesis Agent: Generates clean document drafts tailored to the specific target metrics.

The end-user interacts through a clean, single-page UI built in Streamlit that hides credential management fields for security, letting users focus completely on generating structured institutional insights instantly .

Technology Used

- **LangFlow Platform:** For designing, mapping, and testing visual component connections for the orchestration layers.
- **IBM watsonx.ai Foundation Models:** Large-scale language infrastructure used to run reasoning pipelines and generate contextual syntax instructions.
- **IBM Cloud Environment:** Provides the underlying compute, API credentialing, and project workspace allocations.
- **Streamlit Framework:** Used to construct the lightweight, interactive user dashboard interface.
 - **Retrieval-Augmented Generation (RAG):** Connects external database manual reference lookups to contextual execution prompts.

Application Components Used

Chat Input / Parameter Query Area: Interface where users enter their specific criteria requests or unstructured institutional text blocks.

IBM watsonx AI Foundation Interface Model: Communicates directly with **the meta-llama/llama-3-3-70b-instruct** model endpoint inside the deployment parameters.

Parameter Controls Configuration: Implements explicit greedy decoding controls (**DECODING_METHOD: "greedy"**) and optimized max token bounds to prevent text truncation.

Chat Output / Markdown Presenter: Renders the resulting script text, structured dictionary objects, and validation verification statuses seamlessly for the user.

LangFlow / Application Components Used:

Chat Input/Text Area Component: The front-end text collection mechanism where criteria prompts or schema requests are input by the user.

IBM watsonx AI Foundation Interface Model: Communicates with the meta-llama/llama-3-3-70b-instruct model endpoint inside the deployment parameters.

Parameter Controls Configuration: Implements explicit greedy decoding controls (DECODING_METHOD: "greedy") and strict token bounds for structural accuracy.

Chat Output / Markdown Presenter: Renders the resulting script text, structured JSON objects, and verification statuses seamlessly.

Documents Uploaded:

github.com/trisha-tulabandula/NAAC-accreditation-agent

trisha-tulabandula / NAAC-accreditation-agent

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trisha-tulabandula

Hardcode keys and remove authentication sidebar

76d981b · 11 hours ago

6 Commits

README.md	Initial commit	2 days ago
app.json	Add app.json configuration for NAAC agent	2 days ago
app.py	Hardcode keys and remove authentication sidebar	11 hours ago
langflow_pipeline.json	Add NAAC Accreditation Workflow JSON configuration	yesterday
requirements.txt	Add required packages to requirements.txt	2 days ago

README

NAAC-accreditation-agent

About

No description, website, or topics provided.

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors 1



NAAC Accreditation Multi-Agent Assistant

Powered by Llama 3.3 & watsonx.ai RAG Workflow

Ask the NAAC Agent / Request SSR Generation Draft:

Generate the narrative description draft and structural schema data metrics for NAAC Criteria 1.1.1 regarding curriculum design, academic calendar planning, and tracking annual syllabus updates.

Execute Multi-Agent Synthesis

✓ Analysis Complete

Generated Document / Advisory Input:

NAAC Parameter Mapping Schema Configuration

```
{
  "$schema" : "[https://json-schema.org/draft/2026-03/schema#](https://json-schema.org/draft/2026-03/schema#)"
  "title" : "NAAC_Criteria_Metric_Validation_Map"
  "type" : "object"
  "properties" : {
```



NAAC Accreditation Multi-Agent Assistant

Powered by Llama 3.3 & watsonx.ai RAG Workflow

Ask the NAAC Agent / Request SSR Generation Draft:

Draft a compliance summary template for Criteria 2.2.1 examining catering to student diversity, full-time teacher ratios, and innovative experiential learning methodologies.

Execute Multi-Agent Synthesis

✓ Analysis Complete

Generated Document / Advisory Input:

NAAC Criteria 2.2.1: Student Diversity & Teaching-Learning Process

Narrative Description Draft:

The institution assesses the learning levels of students post-admission and organizes special programs for both slow and advanced learners. Experiential learning, participative learning, and problem-solving methodologies are deeply integrated into departmental lesson profiles. The institutional full-time teacher ratio strictly adheres to statutory council mandates, ensuring continuous personal mentoring.

```
{  
  "criteria": "2.2.1"
```

NAAC Criteria 2.2.1: Student Diversity & Teaching-Learning Process

Narrative Description Draft:

The institution assesses the learning levels of students post-admission and organizes special programs for both slow and advanced learners. Experiential learning, participative learning, and problem-solving methodologies are deeply integrated into departmental lesson profiles. The institutional full-time teacher ratio strictly adheres to statutory council mandates, ensuring continuous personal mentoring.

```
{
  "criteria": "2.2.1"
  "metric_type": "Quantitative Verification Matrix"
  "slow_learner_programs_tracked": true
  "advanced_learner_incentives": [
    0 : "Research Stipends"
    1 : "Peer Mentoring Roles"
  ]
  "student_teacher_ratio": "18:1"
  "status": "COMPLIANT"
}
```

Est. Input Tokens

415

Est. Output Tokens

980

Total Tokens Tracked

1,395

Novelty and Uniqueness

Automated Schema Generation: Transforms manual verification into instantly queryable programmatic configurations (e.g., auto-formatting metrics collections seamlessly).

Zero-Overhead Interface: Uses automated configuration fallbacks to bypass complex manual API keys setups on runtime screens.

Compliance Grounding: Designed specifically to map direct parameters from educational manual databases to text output pipelines.

Future Scope

Autonomous Silo Integration: Connecting backend file-parsers directly to active university storage databases (Google Drive, local servers) to auto-extract text inputs.

Cross-Criteria Compliance Auditing: Expanding the reasoning loop to match real-time text outputs against strict historical grading rubrics from the official council guidelines automatically.

Git Hub Link

Working Github URL –

<https://github.com/trisha-tulabandula/NAAC-accreditation-agent.git>

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Certificates Earned:

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Learning hours: 30 mins

Thank You!

Thank you for your time and interest.